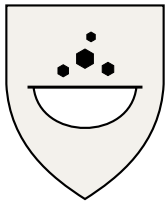
ON CERTAIN PLAGUES ARISING FROM MICRO-MANA

BY

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DEROLA

DEROLA: PRINTED AT THE SIGN OF THE DISH

C.R. 1192

PREFACE

Severin's Twenty-First Question asks: "Whence come the plagues?" For more than three hundred years the physicians of the cities have each had their own answer. Some say the plagues come of foul mana, which in the tide years grows foul and is breathed in by mortals and beasts; others that they pass from the sick to the whole, and so travel along the air-routes from city to city. Both have something in them, and neither tells all: on the first, why of those who live in one city and breathe the same mana do some fall sick and others not; on the second, why in the tide years do many cities that have no dealings with one another fall sick at once?

The year C.R. 1190 was a tide year. From the Month of Reawakening the mortals of Derola, the people of mixed blood and the sheep of the city's outskirts took the tide-carbuncle one after another: first a fever, then after some days black carbuncles under the skin, and in the worst cases death. The Silver Folk of the same city, as always, took it not at all. I watched this plague from the hospital of the city, and made use of it to do the things that follow. I shall prove that the tide-carbuncle, and certain other plagues that can be raised again in beasts, arise from a very small living thing that carries mana and multiplies. I call it micro-mana.

CHAPTER 1

PLAGUE AND TIDE

The hospital of Derola has kept a plague-register since 1151, in which every case and every sick beast is written with its day and its quarter of the city, the people of mixed blood being written among the mortals. I have summed the forty years of the register year by year; the sums are given in Table I, and drawn year by year in Fig. 1.

Table I. The tide-carbuncle in the plague-register of Derola, 1151–1190 (yearly means)

Years	*Number of years*	*Cases among mortals*	*Sick sheep*	*Deaths among mortals*
Tide years (1160, 1170, 1180, 1190)	4	286	1,730	41
Other years	36	23	154	3

In the register the cases among mortals in the tide years are some twelve times those of the other years; and of the cases of the four tide years taken together, seventy-nine in a hundred fall in the three months of Reawakening, Summer Haze and Scrolls, the very months in which the moon of mana stands in the western sky. That the plague rises and falls with the tide others have said before; but the register shows a further thing that has not been remarked: outside the tide years the tide-carbuncle does not die out: every year there are cases, never fewer than twelve nor more than thirty-three, mostly in the quarters along the air-routes.

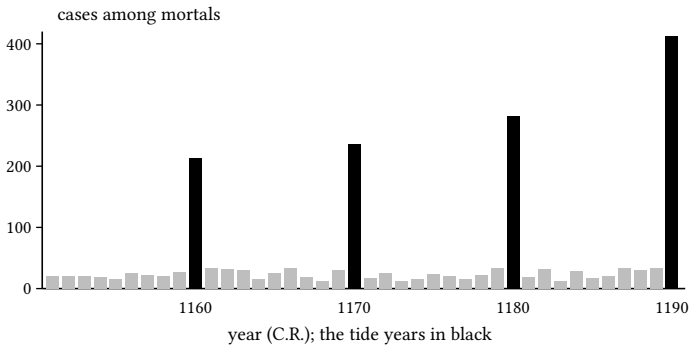


Fig. 1. Cases of the tide-carbuncle among mortals, year by year, in the plague-register of Derola, 1151–1190; the tide years in black

Of this outbreak of 1190 the register is most exact. On the third day of the Month of Reawakening a herdsman of the East Wall brought the first sick sheep; on the fifth, mortals fell into fever on the North Slope and at the West Quay; by the end of the month nine of the eleven quarters of the city had cases. In the three months after, when the moon of mana was not in the western sky, the new cases were far fewer. The Month of Summer Haze was the worst, with more than a hundred and ninety cases among mortals in the one month and more than twelve hundred sick sheep; after the Month of Scrolls the cases grew fewer, and by the end of the year there had been four hundred and twelve cases among mortals, sixty-three deaths, and two thousand six hundred and ten sick sheep. The mortals who fell sick were mostly herdsmen, tanners and the carriers of the quays; of those who lived under one roof, some fell sick and some did not.

CHAPTER 2
SEEING THE MICRO-MANA

If there is anything in a fluid of the body that carries mana, a working for the sensing of mana will feel it; but it is too small to be seen, only felt. I asked the array-maker of the hospital to carve an array of magnifying upon a thin plate of crystal: a drop of fluid is set upon the plate, a working of sensing of the first ring is cast upon it, and the array enlarges the light the working returns and throws it upon a white screen, where the image is four hundred times the size of the thing, the screen being kept always at one distance from the plate. The power of magnifying is fixed thus: several fine crystals of measured sizes, not all alike, are set upon the plate, and their sizes upon the screen are observed; for each of them it comes to some four hundredfold. Measured in this way, a micro-mana is not longer than a tenth of the thickness of a hair.

Not every point of light upon the screen is a living thing. Dust upon the plate, splinters of mana crystal in the fluid, and even glints of the array itself, all throw points of light. I tell them apart by two rules. The first is to follow one and the same point of light: the living divide, and what they divide into divides again, which dust, splinters and glints do not; that a point moves is no proof that it lives, since the fluid may carry it, and I take movement only as a sign beside the rest. The second is to set the plate within an order ward: the living cease to move and to divide within a day, while dust and glints remain as

they were. By these two rules I learned what the living look like; once that was learned, they were known by their look, without waiting for each to divide. In counting, each drop is looked at in ten places, the number upon the screen counted at each, and the mean of the ten taken; this is the mean in a place of Table II, and the portions in which none were seen are reckoned in it.

Looked at in this way, the fluids of the sick show upon the screen many very small points of light, shaped like grains of rice, somewhat long, some joined in pairs, some dividing into two. They move of themselves and avoid one another, and left overnight beneath a sealed plate they double in number. In the fluids of the whole there is not one. They carry mana, they are alive and they multiply; I call them micro-mana. The left of Fig. 2 shows micro-mana as seen upon the screen.

I took the fluids of the sick in a tide year and in another year, of the whole, of sick sheep and of whole sheep, one portion from each man and each beast, and always of the blood; those of the tide year were taken in 1190, those of the other year in 1191, and those of the sheep in 1190. In one drop of each portion I counted the micro-mana; the counts are given in Table II.

Table II. Micro-mana in one drop of fluid, seen at four hundredfold

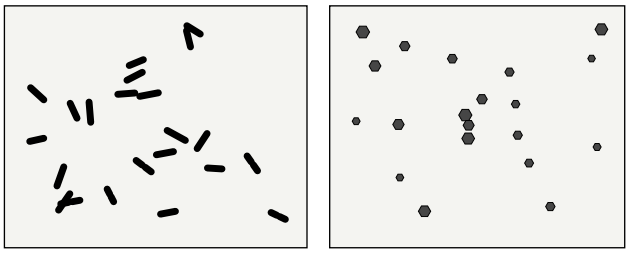
Taken from	*Portions*	*With micro-mana*	*Mean in a place*
The sick, in a tide year	60	57	31
The sick, in another year	40	33	7
Whole mortals	50	0	0
Sick sheep	40	40	44
Whole sheep	30	0	0

In the fluids of the sick there are more than four times as many micro-mana in a tide year as in another; in whole men and sheep, looked at in this way, not one is to be seen. Among the sick there were some in which none were seen, three portions of the sixty in the tide year and seven of the forty in the other; why, I cannot say.

CHAPTER 3
THE REMAINS

Latham's Taxonomy of Monsters says that most monsters leave mana crystals when they die. If the micro-mana are monsters, they too should leave mana crystals when they die. I set apart micro-mana seen under the magnifying, and by the third day the points of light no longer moved, but shrank one by one into very fine grains of crystal, with edges and corners, and with a cold light unlike that of the living; these are the mana crystals of dead micro-mana, shown on the right of Fig. 2.

Accordingly I sought such grains in the bodies of the dead, men and beasts. Of twenty-five mortals dead of the tide-carbuncle, they were found in the carbuncles and the blood of twenty-two; of forty sheep dead of it, in thirty-seven; of thirty whole sheep slaughtered, and of twelve mortals dead of other sicknesses, not a grain. The grains go with the sickness and not with health; in the three mortals and three sheep dead of it in whom none were found, I could not learn why.



living micro-mana crystals of the dead

Fig. 2. Seen under the array of magnifying: left, living micro-mana, some of them dividing; right, the fine mana crystals into which they shrink when dead

CHAPTER 4
KEEPING APART, AND INOCULATION

Seeing alone cannot show that the sickness is caused by them: they may be the effect of the sickness and not its cause. To prove that it is caused by them, they must be kept apart from the sick body and raised elsewhere, until everything else brought out of the sick body is diluted to almost nothing, and then put into whole beasts, to see whether the sickness is raised again.

I use a sealed array-dish: a sealing array upon its mouth lets nothing in from without and nothing out from within, and the dish holds some hundred drops of a fluid for raising them, made from the serum of sheep. The serum is first boiled and strained, put into the dish and sealed, and kept for three days in a warmth near that of a sheep's body; only if in three days not one point of light is seen upon the screen is it used. Every dish that fails this rule I pour away. A drop of blood is taken from a sheep sick in a tide year and put into the dish, and the micro-mana multiply generation after generation. When they have grown thick, in about three days, one drop is put into a new dish and raised again; and so from dish to dish, twenty passages taking two months. At every passage a blank dish was opened in the same way and given one drop in the same way, only of fluid without micro-mana, and raised beside the rest; in the blank dishes of all twenty passages not one point of light was ever seen. To obtain a single kind, a drop from the first dish is diluted until only one micro-mana is seen under the magnifying, all ten places being looked at, and that drop is put whole into a new dish and raised. I obtained a single kind in this way three times, from three different sick sheep, and what the three dishes raised was alike in shape, in size and in the manner of multiplying; the first of them is that used in Table III.

At each passage one drop goes into some hundred drops of new fluid, so that whatever was there before is left at a hundredth; by the tenth passage everything else in the first drop of blood had been diluted a hundredfold ten times over, and none of it could remain. I put micro-mana of the tenth and of the twentieth passage into whole sheep; I put the fluid from the blank dishes, raised alike, into another flock of whole sheep for comparison; and I put the blood of sick sheep, unraised, directly into a third, for reference. The results are given in Table III.

TABLE III. The results of inoculation (sheep)

Group	*What was put in*	*Sheep*	*Fell sick*	*Micro-mana raised again*
A	One kind, tenth passage	12	11	11
B	One kind, twentieth passage	8	7	7
C	Fluid of the blank dishes	12	0	—
D	Tenth passage, warded first	12	0	—
E	Blood of sick sheep, unraised	6	6	6

The sheep inoculated were all taken from a flock outside the city that had never been sick, and their blood was looked at beforehand and showed no micro-mana; each group was kept in its own fold and watched for twelve full days. The manner of inoculation is to prick the skin on the inner side of the sheep's leg and let in one drop; counted as before, a drop of the culture showed some thirty micro-mana in a place. Of the twenty sheep into which raised micro-mana were put, eighteen took the tide-carbuncle, with the same signs as were seen in the city: fever on the second day, a black carbuncle at the place of the prick on the fourth, and from the sixth either death or a slow mending, seven of the eighteen dying; and from every sheep that fell sick the same micro-mana were raised again, alike in shape, in size and in the manner of their multiplying. Of the sheep into which the fluid for raising was put, not one fell sick. In groups A and B one sheep each did not fall sick, nor had any fever in the twelve days, nor showed micro-mana in its fluids; I suspect that the prick was too shallow and little of the drop went in, or that these two sheep do not take it by nature; but I did not search into it, and do not presume to say so, nor did I try again.

The tide-carbuncle of mortals I cannot raise again by inoculating men; but micro-mana taken from the sick and raised in the dish are alike with those of the sheep in shape, in size and in the manner of multiplying, and leave the same grains when dead. I also took a single kind from the blood of two of the sick, one from each, raised each to the tenth passage and put it into four whole sheep; seven of the eight took the tide-carbuncle, and from every one of them the same micro-mana were raised again. And so I take them to be one kind.

CHAPTER 5
THE ORDER WARD

Group D of Table III is the same kind, of the tenth passage, first set for two days within an order ward and then put in. Within the ward the micro-mana divided no more from the first day, and by the second had all shrunk into grains; a drop of it put into a new dish and raised for three days did not multiply again. Of twelve sheep, each given one drop of a culture so treated, not one fell sick: when the micro-mana of the culture are dead, inoculation causes the sickness no more.

I have also tried setting an order ward over a sick sheep. The sheep was not the better for it, nor did the micro-mana in its fluids grow fewer; it may be that a ward covering a whole sheep falls far short of the strength of a ward covering a single dish.

CHAPTER 6
WHENCE THE MICRO-MANA COME

I hold that the micro-mana are very small monsters cast forth by the tide. I have three reasons. First, in the tide years both the cases and the micro-mana in the fluids are several times as many, and mostly in the tide months. Secondly, in the first ten days of the Month of Reawakening of 1190, the first twenty and more cases in the city fell in four quarters far apart, the East Wall, the North Slope, the West Quay and the South Terrace; in those days the harbour book shows no ship come from any stricken city, and the sick had had no dealings with one another. Thirdly, the micro-mana leave mana crystals when they die, according to the rule of monsters; and monsters are what mana, twisted, casts forth into the present world, and in the tide mana rises and falls most, and casts forth most.

Yet I do not deny that the sick and the beasts carry them away. In the register, the cases of each year outside the tide years, a dozen to thirty and more, are mostly in the quarters along the air-routes; and after the Month of Summer Haze of 1190 the new cases in the city were mostly about the quays. Whether the micro-mana are cast forth by the tide where they are found, or brought along the air-routes by the sick and by beasts, or both, and in what measure each, the physicians of other cities must try each for himself by my method: in their own cities to see, to raise and to inoculate, and to compare the cases with the traffic of the routes.

CHAPTER 7
CONCLUSION

From the chapters above I have proved four things. First, the tide-carbuncle rises and falls with the ten-year tide, and in the tide years there are most micro-mana in the fluids of the sick. Secondly, the fine mana crystals into which dead micro-mana shrink are found in the bodies of men and beasts dead of the tide-carbuncle, and not in the whole. Thirdly, micro-mana taken from a sick body and kept apart multiply generation after generation; a single kind put into whole sheep raises the same sickness, and the same micro-mana can be raised again from the sick sheep; the fluid for raising without micro-mana does not do so. Fourthly, after treatment with an order ward the micro-mana multiply no more, and inoculation no longer causes sickness.

I hold that to say that a plague which rises with the tide and can be raised again in beasts arises from a micro-mana, all four must be proved of it; if any is lacking, one can say only that it is suspected. Those so proved are, to this day, the tide-carbuncle and two other plagues, of sheep and of the floating fowl, whose registers, fluids and inoculations are written in the books of the hospital and not printed here; of other plagues I do not yet know whether they can be raised again in beasts, and so I do not presume to say that they too arise from micro-mana.

There are two things more that I cannot answer. Why the Silver Folk do not take these plagues: I have taken the fluids of more than ten of the Silver Folk of the city and seen not one micro-mana, but whether they breathe them in and cast them out, or cannot take them in at all, I do not know. Whether the micro-mana in the sick can be killed without harm to the sick: the ward I tried over a sheep did no good, and some other means must be found.

I have had these chapters printed, and I ask the physicians of every city to come and search my register, to look at my dishes, and to do it all again by my method in their own cities.